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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VII Semester B. E. Regular Examinations Mar/Apr-2025

Information Science and Engineering

VIRTUAL REALITY AND AUGMENTED REALITY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Differentiate between Augmented Reality and Virtual Reality.	02	2	1
	1.2	List the different input devices used for Virtual Reality applications.	02	1	1
	1.3	Write any two applications of Mixed Reality.	02	1	1
	1.4	What are the default methods while working with Unity?	02	2	1
	1.5	Compare Edit Mode and Object Mode in Blender.	02	2	2
	1.6	Mention different types of Sensor Based Tracking in AR.	02	1	1
	1.7	Illustrate the importance of Degree of Freedom in VR.	02	3	3
	1.8	Outline the importance of Box collider and Rigid body in Unity software.	02	2	1
	1.9	List any four $3D$ modeling tools.	02	1	1
	1.10	List the 4 key elements in experiencing virtual reality.	02	1	1

PART-B

2	a	List out the application of Virtual Reality in different fields and its benefits in the field of education.	06	1	2
	b	Explain the importance of aural representation in VR and how it is used to achieve a complete VR experience.	06	2	1
	c	Discuss the important features of Unity that makes it a popular game engine.	04	2	2
3	a	Write a program for smooth movement of the object along the movement of arrow keys.	10	3	3
	b	Analyze the importance of yaw, pitch and roll in $3D$ rotation.	06	3	4
		OR			
4	a	Illustrate the matrix multiplication algorithm iterated over the chain of matrices to produce “Chaining the Transformations”.	10	3	3
	b	Analyze the importance of triangle meshes for virtual objects.	06	3	4
5	a	Differentiate between First Person Controller and Third Person Controller.	08	3	3
	b	Analyze the need for $3D$ software in Unity. Explain Blender software along with its important features.	08	4	4
		OR			
6	a	Illustrate the need for Sclupting and Rigging features in Blender.	08	3	3

	b	Consider a shooting game which has to be developed. Analyze its advantages and disadvantages if developed using First Person Controller and Third Person Controller.	08	3	4
7	a	Explain the advantage of Mobile Sensors in Tracking.	08	2	1
	b	Identify the problems which has to be solved to make an effective tracking system.	08	2	2
		OR			
8	a	Explain Optical Tracking in detail along with its advantages and disadvantages.	08	2	1
	b	Differentiate between Marker-based and Marker-less tracking along with its application.	08	2	2
9	a	Explain WebXR and the challenges associated with it.	08	2	1
	b	Analyze the different ways in which the VR/AR session may end and suggest the appropriate steps to be followed when the session has to end.	08	3	4
		OR			
10	a	Explain the Life cycle of WebXR application in detail.	08	2	1
	b	Explain the functionality for WebXR request and session life cycle.	08	2	4